SC Honors Geometry - Nappier
6.2 – Bisectors of Triangles • Homework

Name: ____________________________________

Answer each question about points of concurrency.

1. The circumcenter of a triangle is equidistant from what parts of the triangle? The incenter?

2. Where does the circumcenter of a right triangle lie? Be specific.

3. Where does the circumcenter of an obtuse triangle lie? Where does the incenter of ANY triangle lie?

Find the circumcenter.

4. $\triangle ABC$ with $A(0, 0)$, $B(0, 8)$, and $C(6, 0)$. Give the circumcenter and its distance to each vertex.

5. $\triangle DEF$ with $D(-3, 5)$, $E(-3, -1)$, and $F(5, -1)$. Give the circumcenter and the common distance.

Solve.

6. P is the circumcenter of $\triangle XYZ$, $PX = 3a + 4$, and $PY = 5a - 6$. Find a and PZ.

7. Q is the incenter of $\triangle RST$. Its distance to $\overline{RS}$ is $2b + 7$ and its distance to $\overline{ST}$ is $4b - 3$. Find b and the distance from Q to $\overline{RT}$.

8. The incenter of $\triangle ABC$ is 6 cm from each side. What is the radius of the largest circle that fits inside the triangle?

Decide which point of concurrency each situation calls for. Explain.

9. A fountain must be the same distance from the three straight paths that bound a triangular park.

10. A well must be the same distance from three houses at the corners of a triangular lot. What happens if the lot is obtuse?